

THREE MINI LESSONS



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Centering adjustment

We have three options for centering in HLM

- Uncentered (original metric)
- Grand-mean centered
- Group-mean centered

The first two options will include an adjustment to coefficients for group-level differences in the predictor.

Group-mean centering adjusts for these differences by considering deviations from the group mean. The coefficient is therefore unadjusted.

Centering adjustment

Consider an model with an uncentered predictor

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij}$$

If X_{ij} is not centered, it can be decomposed into:

$$X_{ij} = (X_{ij} - \bar{X}_j) + \bar{X}_j$$

Substituting into the model

$$Y_{ij} = \beta_{0j} + \beta_{1j}(X_{ij} - \bar{X}_j) + \beta_{1j}\bar{X}_j + r_{ij}$$

Centering adjustment

The coefficient is therefore a blend of the within-group effects and between-group effects.

The adjustment will depend on how much variation there is in the covariate between groups.

Fortunately, in most cross-sectional and longitudinal applications, this adjustment isn't large.

Unit Specific versus Pop Av

HGLMs with non-linear link functions report two different sets of estimates

- Unit specific
- Population average

These address different research questions

Unit Specific versus Pop Av

The key reason for this is that it matters how you convert your model-based estimates back into probability.

$$E[f(X\beta + u)] \neq f(E[X\beta + u])$$

Unit Specific versus Pop Av

In the case of the logistic link function

$$f(\eta) = \frac{e^{\eta}}{1 + e^{\eta}}$$

$$E \left[\frac{e^{X\beta+u}}{1 + e^{X\beta+u}} \right] \neq \frac{e^{E[X\beta+u]}}{1 + e^{E[X\beta+u]}}$$

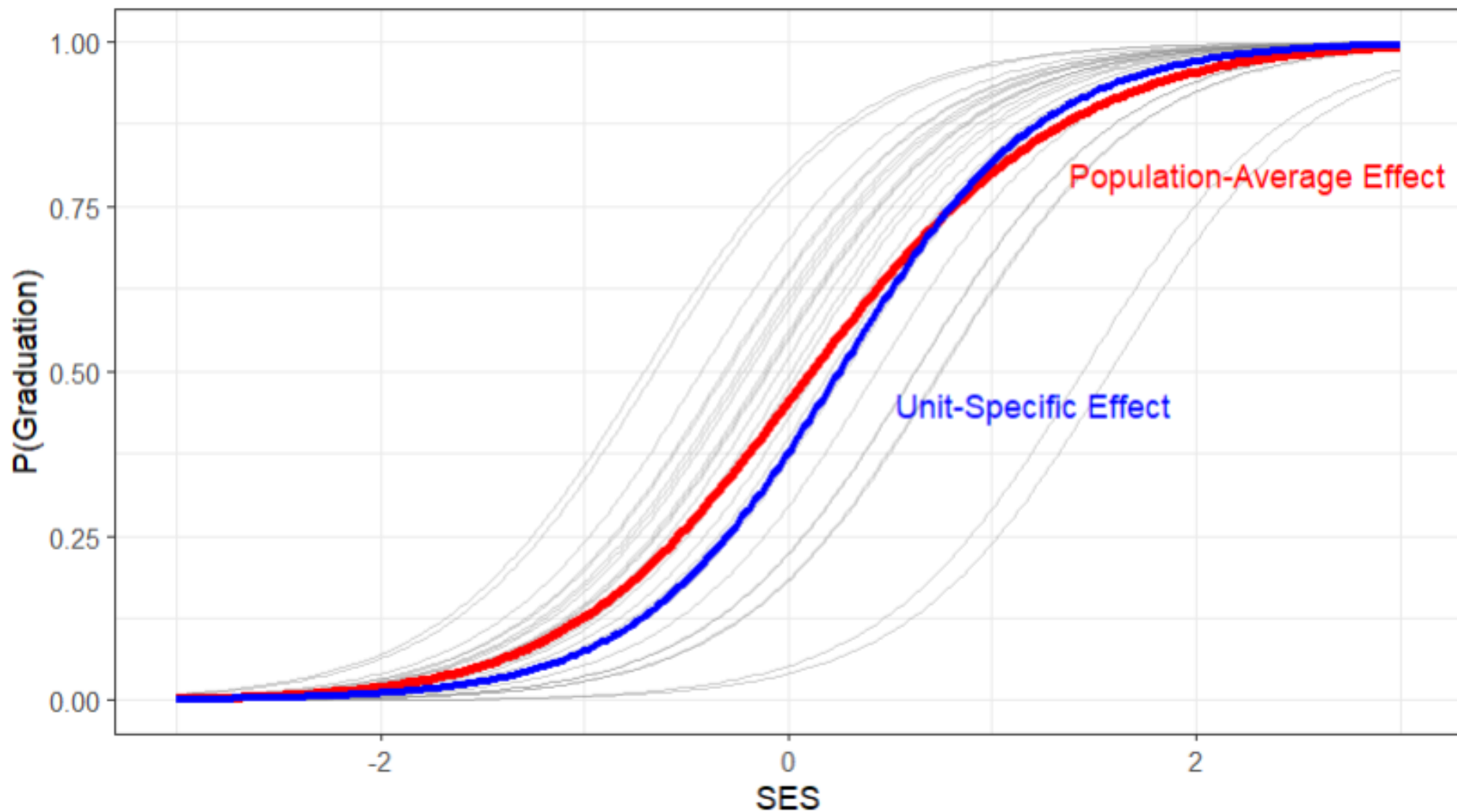
Unit Specific versus Pop Av

So we have two options:

- Convert into a probability first, and then average.
 - This yields population average results
 - But because some coefficients will have higher or lower intercepts, the associated changes in probability will not be the same
- Ignore the variation (set $u = 0$), then convert
 - This yields unit-specific estimates

Unit-Specific vs Population-Average Effects of SES on Graduation

Why population-average effects are smaller in multilevel logistic models



Missing data

When we talk about missing data, we generally mean missing values for certain variables in observed cases.

This is distinct from missing cases. You will always have missing cases unless your data represent a population census. You are aware that non-random selection into your sample can bias your estimates. Similarly, omitting cases with non-random missing data can also bias your estimates.

Missing cases

There are some circumstances where missing cases are considered missing data – such as missing observations in a time series analysis.

HLM is generally robust to missing cases in terms of providing efficient estimates.

There are concerns that non-random selection of cases, especially in longitudinal models (e.g. non-random attrition), can bias your results.

Missing Data

There are three types of missing data:

- MCAR – Missing Completely At Random
- MAR – Missing at Random
- MNAR – Missing Not at Random

MCAR

Missing Completely At Random

- No relationship between missingness and other non-missing data.
- Missing data are a simple random sample of complete data
- Nothing systematic going on
- Poses no threat to statistical inference (i.e. introduces no bias, but may reduce sample size in complete case analysis)

MAR

Missing At Random

- There is a relationship that among the missing values, but it can be **completely** accounted for by observed data
- For example, if men are more likely to report their weight or age than women, and we know gender, then data are MAR
- Deleting observations can lead to bias

MNAR

Missing Not At Random

- Also called *non-ignorable missingness*
- Relationship between missing data and observed data exist but are unknown
- Poses greatest threat to statistical inference
- Most common situation in non-experimental studies. Even in experimental studies, can occur if you don't have 100% compliance with experimental protocols

How to Handle Missing Data

These methods are listed from simplest to most sophisticated. More sophisticated methods are better at reducing bias, but require more work.

- Deletion (listwise or pairwise)
- Constant imputation (with or without a missing indicator)
- Model-based imputation
- Multiple imputation

Listwise Deletion

- Omits data that are missing on any variable that will be considered at any point in any analysis
- Also called *complete case analysis*
- Consistent samples between analyses
- Implicitly assumes data are MCAR. If not, then results will be biased proportional to amount of missing data and strength of the relationship between missingness and observed values.
- If data are MCAR, means (inc. regression coefficients) will be unbiased, but standard errors may be overestimated.

Pairwise Deletion

- Omits data that are missing on any variable in each specific model
- Default for many statistical tests in most packages
- Can result in different sample samples for different analyses
- Implicitly assumes data are MCAR. If not, then results will be biased proportional to amount of missing data and strength of the relationship between missingness and observed values.
- If data are MCAR, means (inc. regression coefficients) will be unbiased, but standard errors may be overestimated.

Constant imputation

- Replace missing values with fixed constant
- Popular choices included zero (definitely include missing flag), the mean or median (continuous variables), or mode (categorical)
- Can include a missing indicator variable in the model as well.
- Will reduce bias if data are MCAR or MAR. Variances and covariances will be misestimated. This will inflate p-values.
- Quick and dirty fix – okay for exploratory analysis but not recommended for important work

Model-based imputation

- Uses regression models or something similar (e.g. maximum likelihood estimation) to predict missing values based observed data
- Reduces bias if data are MAR
- Preserves relationships between variables
- Still mis-estimates variances (hence standard errors)

Multiple Imputation

- Uses models to generate multiple likely values for missing data, which are dispersed into multiple newly-generated copies of the observed dataset
- The final analytic model is run on each dataset, and the results compared
- The final point estimates are averages of these results, and the standard errors are adjusted for variance between them
- Preserves the relationship between variables, gives best estimates of variances
- Results will still be biased if data are MNAR

Multiple Imputation

Tips for MI:

- Include all available data (cleaned for regression analysis) that you can in your imputations
- But.... MI is computationally expensive. Be prepared for it to take time for large datasets and especially with a large number of variables
- Include your treatment and outcome variables as well

Which to choose?

How problematic your missing data is depends on:

- The amount of missingness
- The non-randomness of the missingness

Conduct exploratory analysis to investigate these points.

Use theory to speculate on what might influence missingness in your data.

Which to choose?

If you have a small amount of missing data ($<10\%$), then you can choose one of the more basic techniques.

If you have a larger amount of missing data, you should use a more sophisticated approach.

If you have a very large amount of missing data ($>40\%$) then even sophisticated techniques are insufficient if missingness is MNAR.

Missing Data in HLM

The usual approaches to missing data do not account for the clustering.

This means that contextual effects tend to be lost in the imputation process, introducing bias into results based on imputation.

Shin and Raudenbush (2013) developed a method of doing it correctly that is integrated into the HLM software. See Chapter 20 of the HLM Software User Manual.